

Exam Date & Time: 20-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

INTRODUCTION TO MACHINE LEARNING [NPT-4004]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain DBSCAN Clustering?
Consider the following 2D dataset with 10 points: (1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (8, 8), (8, 9), (9, 8), (9, 9), (10, 10). Analyze the given data using DBSCAN with $\epsilon = 1.5$ and $\text{MinPts} = 3$, find the total cluster groups formed for this dataset? (5)
- A)

B)

A hospital wants to classify patients as either having a heart condition (class 1) or not having a heart condition (class 0) based on two features: Age and Cholesterol Level. The hospital decides to use the K-Nearest Neighbors (KNN) algorithm with $k=3$. The following table represents the data of 6 existing patients, where each row shows the age, cholesterol level, and whether the patient has a heart condition:

Patient	Age	Cholesterol Level	Heart Condition(0 = No, 1 = Yes)
A	45	200	1
B	50	220	1
C	40	180	0
D	35	170	0
E	60	240	1
F	55	230	1

(5)

A new patient arrives with an Age of 42 and Cholesterol Level of 190. Using Euclidean distance, classify the new patient using the KNN algorithm with $k=3$. Show your calculations for the distances and the final classification result.

- 2) (5)

A)

A real estate company fixed the price of each plot as per the given table

Square Footage (sqft)	Number of Bedrooms	Price
1500	3	250000
1800	4	300000
2200	3	350000
1200	2	200000
2500	5	400000

Apply linear regression model and find the regression line and predict the price for 2300 square foot flat with 3 bedrooms.

- B) A medical researcher is working on a project to classify patients into two categories: healthy and diseased, based on a set of medical test results. The dataset contains many features, making it computationally expensive to train a classifier. How can Linear Discriminant Analysis (LDA) be used to address the dimensionality reduction challenge in this scenario? Discuss the potential benefits and limitations of using LDA for this task. What are the key assumptions of LDA, and how can these assumptions impact the performance of the classifier? (5)
- 3) Aanalyze the role of kernel functions in Support Vector Machines (SVMs). Discuss the trade-offs between different kernel functions in terms of model complexity, computational cost, and generalization performance. Provide real-world examples to illustrate the application of SVM with various kernel functions. (5)
- A)
- B) How would you apply the backpropagation technique to optimize the learning process in an artificial neural network? Explain with a neat diagram and suitable example (5)
- 4) (5)
- A)

Consider the following dataset:

price	maintenance	capacity	airbag	profitable
low	low	2	no	yes
low	med	4	yes	no
low	low	4	no	yes
low	high	4	no	no
med	med	4	no	no
med	med	4	yes	yes
med	high	2	yes	no
med	high	5	no	yes
high	med	4	yes	yes
high	high	2	yes	no
high	high	5	yes	yes

suppose you decided to construct a decision tree using binary splits and the Gini index impurity measure.

- (a) price - {low, med}||{high}
- (b) maintenance - {high}||{med, low}
- (c) maintenance - {high, med}||{low}
- (d) capacity - {2}||{4, 5}

Consider the given features and split point combinations which would be the best to use as the root node assuming that we consider each of the input features to be unordered?

- B) How would you design an ensemble learning approach using both bagging and boosting techniques to improve the performance of a machine learning model for a complex classification task? Justify the selection of each technique and discuss how they could be integrated effectively to address bias, variance, and overfitting issues. (5)

- 5) How does the Naïve Bayes classification model work?

- A) Suppose we have a small dataset to predict whether a student will pass or fail a test based on the hours they study. The features are Study Hours (with values: Low, High), and the class is whether they Pass or Fail.

Study Hours	Result
Low	Fail
Low	Fail
High	Pass
High	Pass
High	Pass

(5)

Apply Naïve Bayes theorem and predict whether a student who studies High will pass or fail.

- B) Explain the core concepts of Hidden Markov Models (HMMs), including hidden states, observed emissions, and the transition and emission probabilities. How do HMMs differ from other statistical models like Markov Chains? Provide a real-world example to illustrate the application of HMMs. (5)

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